

No books, notes, computational devices, etc. are allowed. Use clear handwriting and justify your answers carefully. Please write your anonymous identification number on each sheet of paper.

1. Suppose R is a principal ideal domain. Let M be an R -module and let N be a submodule of M . The *isolator* of N in M is

$$\text{Iso}_M(N) = \{x \in M \mid \exists r \in R \setminus \{0\} \text{ such that } rx \in N\}.$$

Show that $\text{Iso}_M(N)$ is a submodule of M , and show further that every non-zero element in $M/\text{Iso}_M(N)$ is torsion-free. Finally, find an example of a proper non-trivial submodule that equals its isolator; i.e. $N = \text{Iso}_M(N)$.

Solution. (5 marks) Write $I = \text{Iso}_M(N)$. If $y, z \in I$, then there are non-zero $r, s \in R$ such that $ry, sz \in N$. Hence $rs(y - z) = s(ry) - r(sz) \in N$ as N is an R -submodule. Also for $t \in R$, we have $r(ty) = t(ry) \in N$, since R is commutative. So I is an R -submodule.

Now let $m + I \in M/I$. If $r(m + I) \in I$ for $r \in R \setminus \{0\}$, then $rm \in I$ and so $m \in I$. Hence R/I is torsion-free.

For the example, let $R = \mathbb{Z}$ and $M = \mathbb{Z} \times \mathbb{Z}$. Then $N = \{(z, z) \mid z \in \mathbb{Z}\}$ equals its isolator, as the isolator consists of all diagonal elements, which is N .

2. Find the rational canonical form of the following matrix over $\mathbb{Q}$:

$$\begin{pmatrix} 2 & 4 & 0 \\ 1 & 4 & 1 \\ 3 & 0 & 3 \end{pmatrix}$$

Solution. Here

$$\begin{aligned}
\begin{pmatrix} 2-x & 4 & 0 \\ 1 & 4-x & 1 \\ 3 & 0 & 3-x \end{pmatrix} &\sim \begin{pmatrix} 1 & 4-x & 1 \\ 2-x & 4 & 0 \\ 3 & 0 & 3-x \end{pmatrix} \quad (R_1 \leftrightarrow R_2) \\
&\sim \begin{pmatrix} 1 & 4-x & 0 \\ 2-x & 4 & x-2 \\ 3 & 0 & -x \end{pmatrix} \quad (C_3 - C_1) \\
&\sim \begin{pmatrix} 1 & 0 & 0 \\ 2-x & 4-(x^2-6x+8) & x-2 \\ 3 & -3(4-x) & -x \end{pmatrix} \quad (C_2 - (4-x)C_1) \\
&\sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & -x^2+6x-4 & x-2 \\ 0 & 3x-12 & -x \end{pmatrix} \quad (R_2 - (2-x)R_1, R_3 - 3R_1) \\
&\sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & x-2 & -x^2+6x-4 \\ 0 & -x & 3x-12 \end{pmatrix} \quad (C_2 \leftrightarrow C_3) \\
&\sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & -x & 3x-12 \\ 0 & x-2 & -x^2+6x-4 \end{pmatrix} \quad (R_2 \leftrightarrow R_3) \\
&\sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & -x & -12 \\ 0 & x-2 & -x^2+9x-10 \end{pmatrix} \quad (C_3 + 3C_2) \\
&\sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & -x & -12 \\ 0 & -2 & -x^2+9x-22 \end{pmatrix} \quad (R_3 + R_2) \\
&\sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 12 & x \\ 0 & x^2-9x+22 & 2 \end{pmatrix} \quad (-C_3 \leftrightarrow -C_2) \\
&\sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{x}{12} \\ 0 & x^2-9x+22 & 2 \end{pmatrix} \quad (\frac{1}{12}R_2) \\
&\sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & x^2-9x+22 & 2-\frac{x}{12}(x^2-9x+22) \end{pmatrix} \quad (C_3 - \frac{x}{12}C_2) \\
&\sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & x^3-9x^2+22x-24 \end{pmatrix} \quad (R_3 - (x^2-9x+22)R_2, 12R_3).
\end{aligned}$$

So the rational canonical form is $\begin{pmatrix} 0 & 0 & 24 \\ 1 & 0 & -22 \\ 0 & 1 & 9 \end{pmatrix}$.

3. Show that there are $\frac{1}{2}(k^n + k^{\lfloor (n+1)/2 \rfloor})$ neckties having n vertical stripes (of equal width), where each stripe can have one of k distinct colours. [The colour of each stripe can be seen on both sides of the tie.]

Solution. (5 marks) Let X be the set of all possible necktie colourings. Clearly $|X| = k^n$. Let $G = \langle a \rangle$ be the cyclic group of order 2. Then X is a G -set under the action of flipping the tie along its vertical (i.e. long) axis.

Clearly the action of any fixed element $a^i \in G$ on a given necktie yields the same necktie. So the number of orbits is the same as the number of different neckties. We now compute X_g for each $g \in G$. First

$$X_e = \{x \in X \mid e \cdot x = x\} = X.$$

So $|X_e| = k^n$. Next

$$X_a = \{x \in X \mid a \cdot x = x\}$$

consists of neckties with the same colour in positions 1 and n , and the same colour in positions 2 and $n-1$, etc. So it suffices to determine the number of patterns in positions $1, 2, \dots, \lfloor \frac{n+1}{2} \rfloor$. Thus $|X_a| = k^{\lfloor (n+1)/2 \rfloor}$.

The result then follows by Burnside's Lemma.

4. Let G be a finite group and $H \trianglelefteq G$ a normal subgroup. Let P be a Sylow p -subgroup of G , for some prime p . Show that $H \cap P$ is either $\{e\}$ or a Sylow p -subgroup of H (or both). Prove also that HP/H is a Sylow p -subgroup of G/H .

Solution. (5 marks) Suppose $H \cap P \neq \{e\}$, else both results are clear. Clearly $H \cap P$ is a p -subgroup. We now show that $H \cap P$ is a Sylow p -subgroup of H . Let K be a Sylow p -subgroup of H that contains $H \cap P$, and let L be a Sylow p -subgroup of G containing K . From the Second Sylow Theorem, we have $P = gLg^{-1}$ for some $g \in G$. Then

$$H \cap P = gHg^{-1} \cap gLg^{-1} \supseteq gHg^{-1} \cap gKg^{-1} = g(H \cap K)g^{-1} = gKg^{-1}.$$

As $H \cap P \subseteq K$, we have $H \cap P = K$, as required.

We now argue that HP/H is a Sylow p -subgroup of G/H . Suppose $|G| = p^r m$ and $|H| = p^a n$, where $\gcd(p, n) = \gcd(p, m) = 1$. By Lagrange's Theorem, we have $a \leq r$ and $|G/H| = p^{r-a} d$ where $d = m/n$.

Now $|HP/H| = |P/H \cap P|$. By the first part, $H \cap P$ is a Sylow p -subgroup of H , so $|H \cap P| = p^a$. This shows that $|HP/H| = |P/H \cap P| = p^{r-a}$, as required.

5. (a) Let $R = M_n(D)$ be the matrix ring over a division ring D , and for $1 \leq i, k \leq n$, denote by E_{ik} the matrix with 1 at the (i, k) entry and 0 elsewhere. Show that $M = RE_{kk}$ is a simple R -module.
- (b) Give an example of a ring R and an R -module M such that M is not completely reducible.

Solution. (5 marks) (a) Let $N \neq 0$ be an R -submodule of RE_{kk} . Note that M is the set of matrices all of whose entries are zero, except possibly in the k th column. Hence a non-zero element of N can be expressed as $\sum_{i=1}^n a_{ik} E_{ik}$ for $a_{ik} \in D$.

Consider $0 \neq \sum_{i=1}^n a_{ik} E_{ik} \in N$. Then for some ℓ , with $1 \leq \ell \leq n$, we have $a_{\ell k} \neq 0$. So $a_{\ell k}^{-1} \in D$ exists. This gives

$$E_{kk} = (a_{\ell k}^{-1} E_{k\ell}) \left(\sum_{i=1}^n a_{ik} E_{ik} \right) \in N,$$

as

$$E_{k\ell} E_{ik} = \begin{cases} 0 & \text{if } i \neq \ell, \\ E_{kk} & \text{if } i = \ell. \end{cases}$$

Hence $RE_{kk} \subseteq N \subseteq RE_{kk}$ giving $N = RE_{kk}$. This proves that M is simple.

(b) Take $R = \mathbb{Z}$ and $M = \mathbb{Z}/4\mathbb{Z}$. Then $N = 2\mathbb{Z}/4\mathbb{Z}$ is a submodule of M , but N has no direct complement P such that $M = N \oplus P$, because this would mean $\mathbb{Z}/4\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$, which is false. Hence M is not completely reducible.

6. Let R be a ring with unity and with the property that every non-empty set of left ideals of R has a maximal element. If R has no non-zero nilpotent ideals, show that R has no non-zero nil ideals.

Solution. (5 marks) Suppose on the contrary that $N \neq 0$ is a nil ideal in R . Let

$$\mathcal{F} = \{\text{Ann}_R(n) \mid n \in N \setminus \{0\}\}$$

be a family of left annihilator ideals. By assumption, there exists $m \in N \setminus \{0\}$ such that $\text{Ann}_R(m)$ is a maximal element in $\mathcal{F}$.

Let $x \in R$. Then $mx \in N$, so there exists a smallest $k \in \mathbb{N}$ such that $(mx)^k = 0$. We claim that $mxm = 0$, and assuming the claim, we then get $(RmR)^2 = R(mRRm)R = 0$. Therefore, by hypothesis, we obtain $RmR = 0$. Then $m = 1m1 = 0$, a contradiction. So $N = 0$.

It remains to establish the claim. Indeed, if $k = 1$, then $mx = 0$ gives $mxm = 0$. If $k \geq 2$, observe that

$$\text{Ann}_R(m) \subseteq \text{Ann}_R((mx)^{k-1}).$$

Since $(mx)^{k-1} \neq 0$, we obtain $\text{Ann}_R((mx)^{k-1}) \in \mathcal{F}$. Then by the maximality of $\text{Ann}_R(m)$, we have

$$\text{Ann}_R(m) = \text{Ann}_R((mx)^{k-1}).$$

Now

$$(mx)^k = 0 \quad \implies \quad mx \in \text{Ann}_R((mx)^{k-1}) = \text{Ann}_R(m) \quad \implies \quad mxm = 0,$$

as claimed.